

Messier: A High-Resolution Corpus for Cross-Benchmark Agent Evaluation



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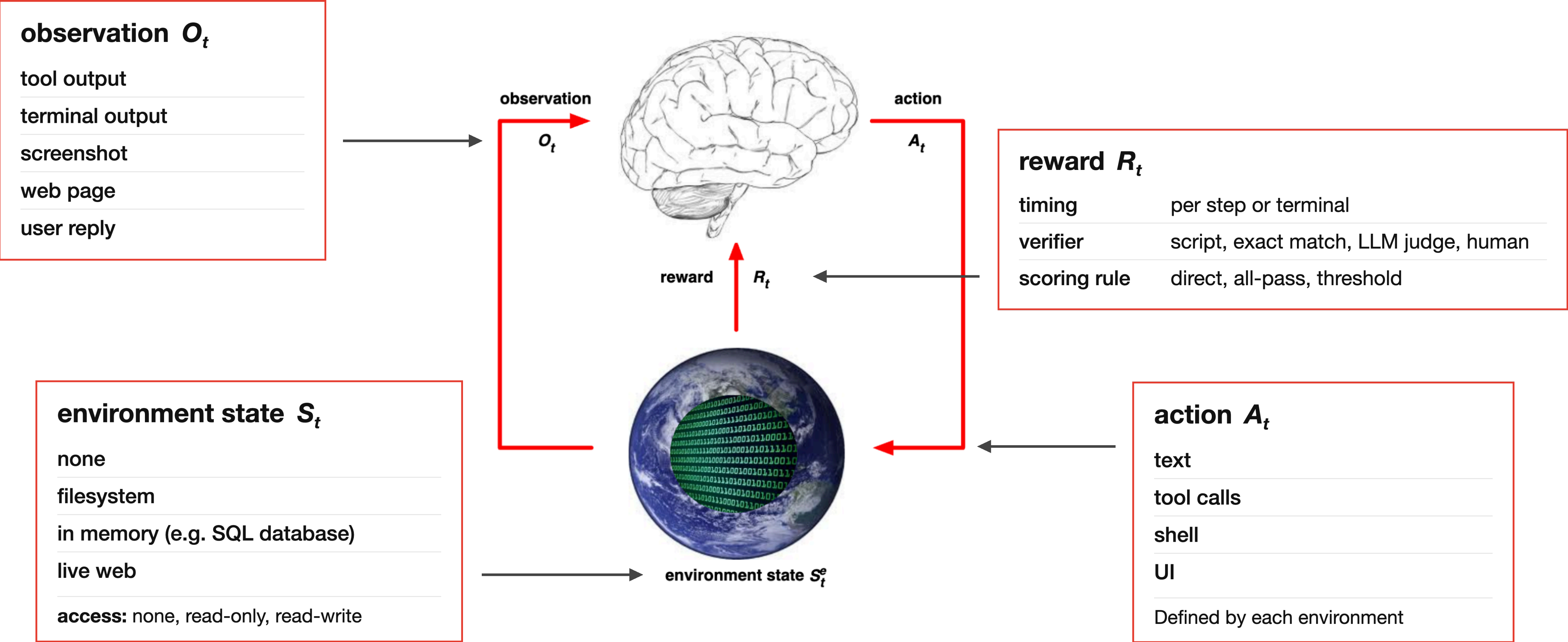
Introduction

Agent benchmarks

- Instrument for measuring how language models act in environments
- Offer coverage of many domains: health, legal, financial, scientific
- Usually defined as a POMDP: state, actions, observations, reward
- **There is no single standard in defining these four components at creation/generation time —> this causes divergence**

Introduction

Environment generation



Introduction

Similar efforts and positioning

- HAL: third-party leaderboard for evaluating agents
→ runs fresh evaluations and does not focus on standardizing existing results



- Epoch AI (ECI): combines scores from many different AI benchmarks into a single scale
→ focuses on software, math and cyber, and releases per-benchmark aggregates only



- METR: measures agent capacity by task horizon length
→ needs human baselines, therefore it covers only 170 tasks



- Arena AI: crowdsourcing + A/B testing
→ answers "which model is best?" through live human comparisons



Introduction

Comparison

Resource	Agentic	Benchmarks	Environments	Models	Scaffolds	Tasks
HELM (Liang et al., 2022)	✗	42	N/A	30	N/A	30,000
METR (Kwa et al., 2026)	✓	3	3	13	N/A	170
Epoch ECI (Ho et al., 2025)	✓	37	N/A	200	N/A	N/A
TOUCAN (Xu et al., 2025)	✓	N/A	495	3	2	1.5M
HAL (Kapoor et al., 2026)	✓	9	N/A	9	~10	~1,500
BRIDGE (Liu et al., 2026)	✓	4	N/A	176	~70	752
Agent Psych. (Ge et al., 2026)	✓	4	N/A	275	~90	1,421
MESSIER	✓	30	2,495	343	221	11,891

Background

Terminology and evaluation flow

- We define a shared vocabulary across benchmarks
- Agent = model + scaffold (i.e. the brain)
- Every task is typed along the axes from before: state, actions, verifiers
- Verifier results are categorized by time, type and way of scoring (aggregation), saving as many bits of information

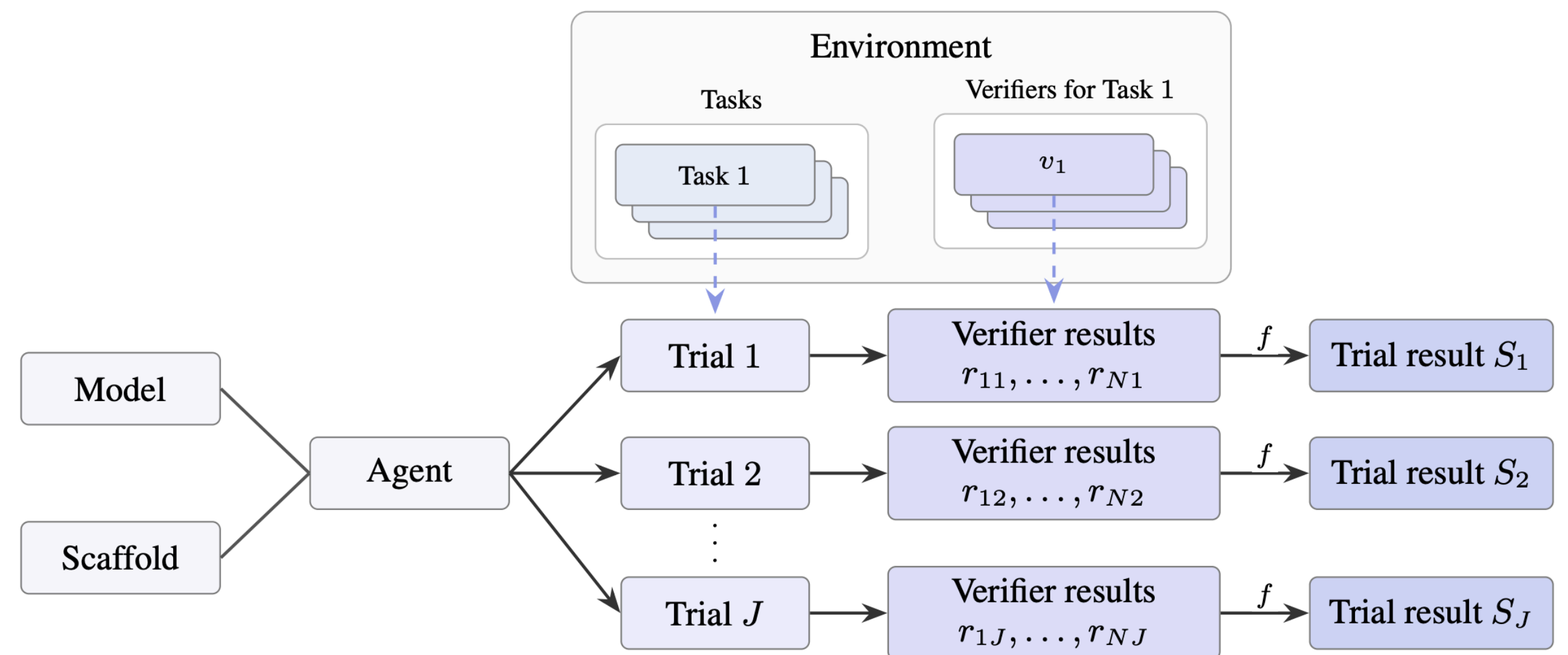
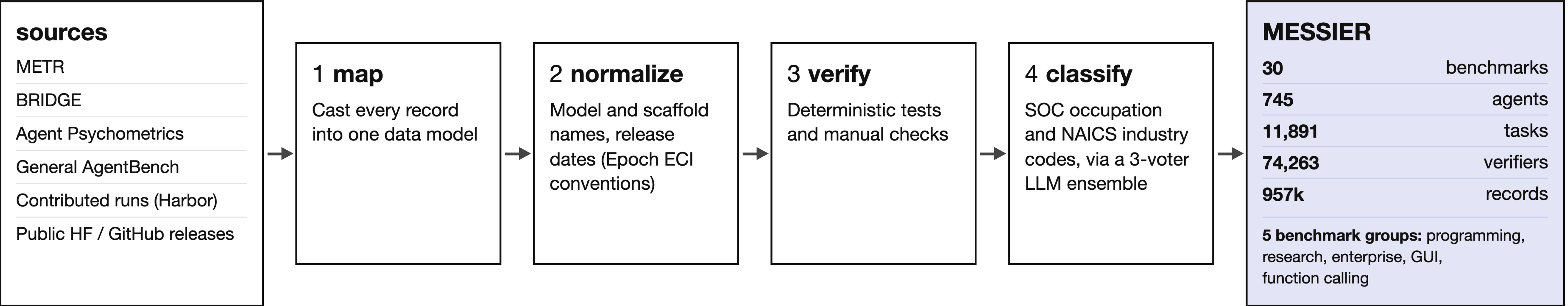


Fig. 1. Benchmark evaluation flow. An environment supports one or more tasks, each associated with one or more verifiers. A model and scaffold form an agent, which may execute the same task over repeated trials. The verifiers evaluate each trial, and the scoring rule combines their outputs into one trial result. Benchmarks usually report only the final result. Meaningful comparison also requires the individual verifier results and the rule used to combine them, so researchers can identify what contributes to success or failure more precisely.

MESSIER

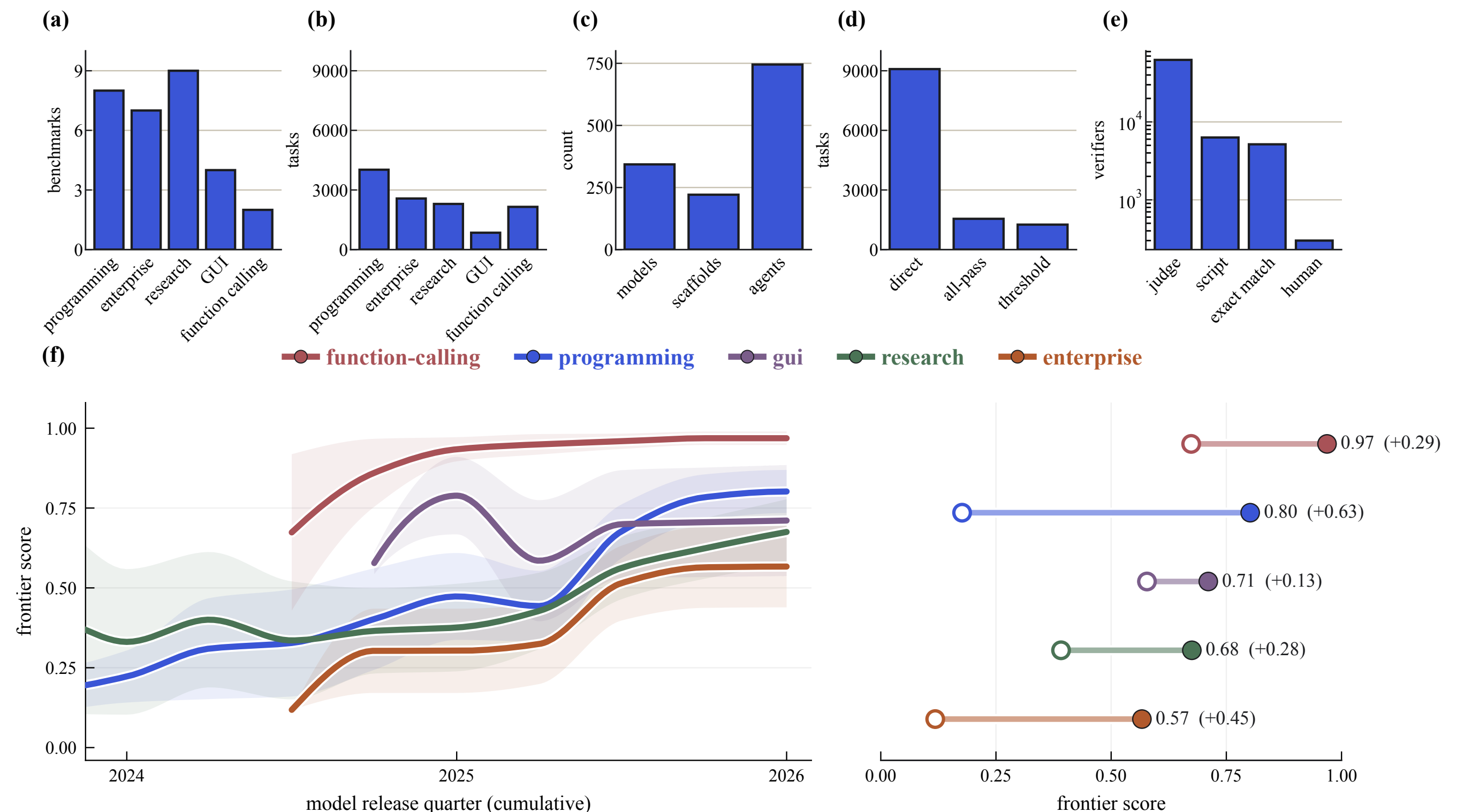
Reconciliation pipeline



Analysis & Applications

Frontier progress

- **Setting:** 30 benchmarks in 5 groups, tracked by model release quarter (2024–2026)
- **Method:** best success rate among agents released so far: averaged per benchmark, then per group
- **Result:** function calling 0.97, programming 0.80 (+0.63, fastest), enterprise 0.57
- **Takeaway:** progress is uneven, where enterprise workflows remain hardest



Note: the groups were qualitatively chosen, thus, there may be some overlap!

Analysis & Applications

Counterfactual rescoring

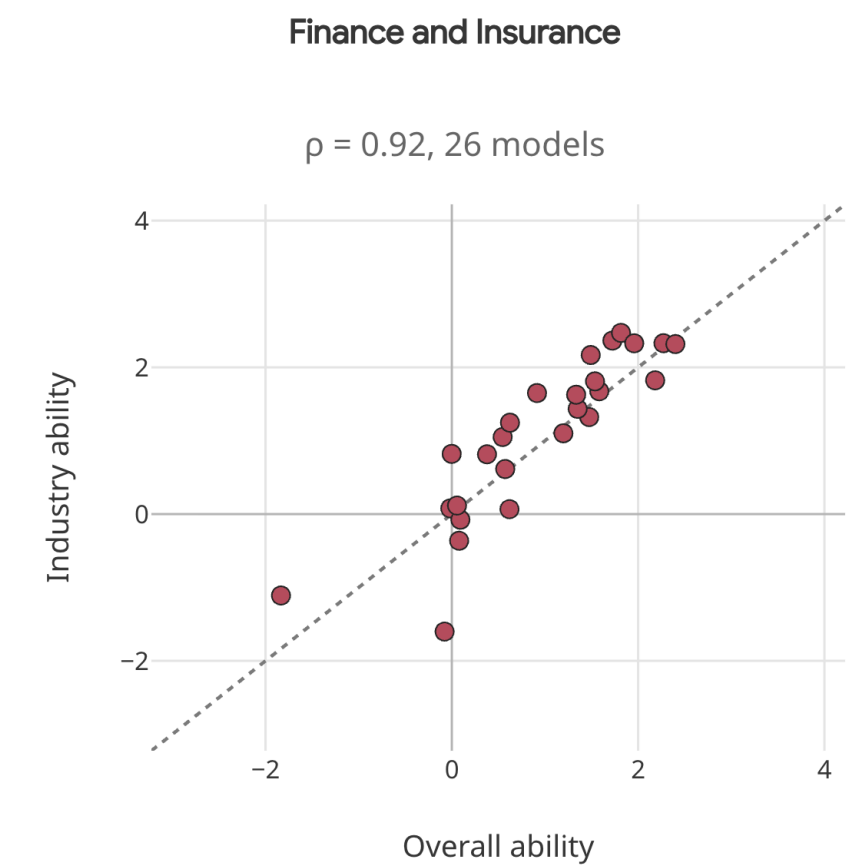
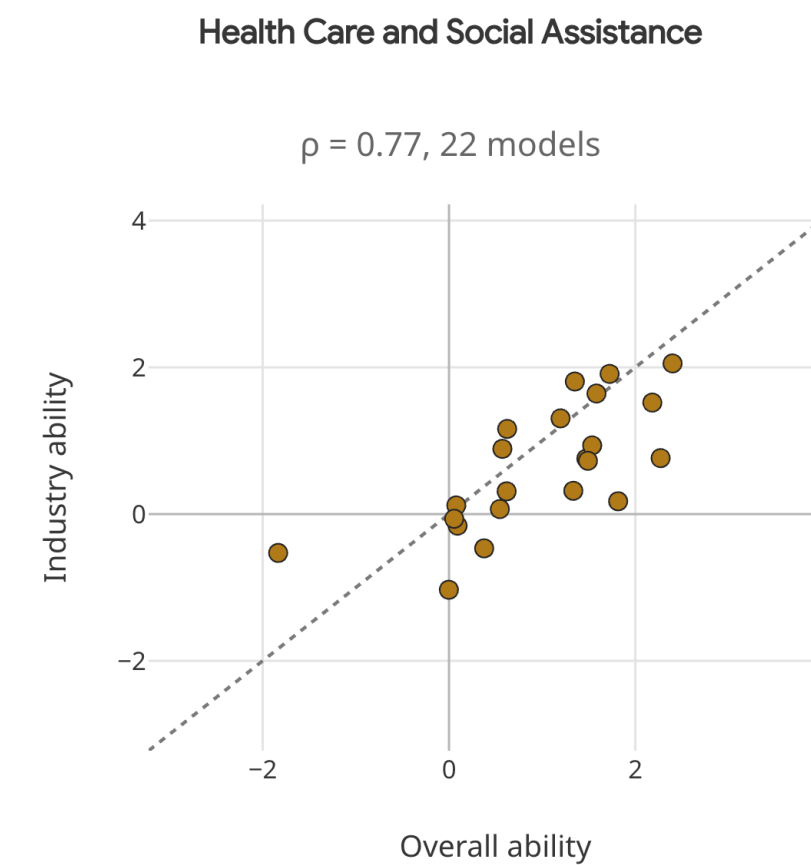
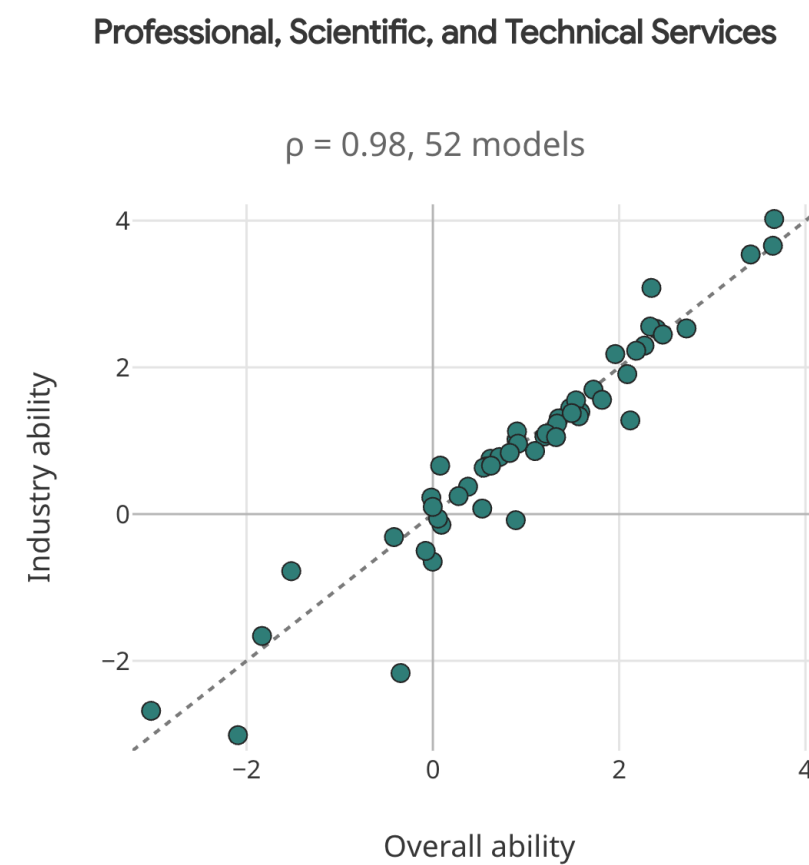
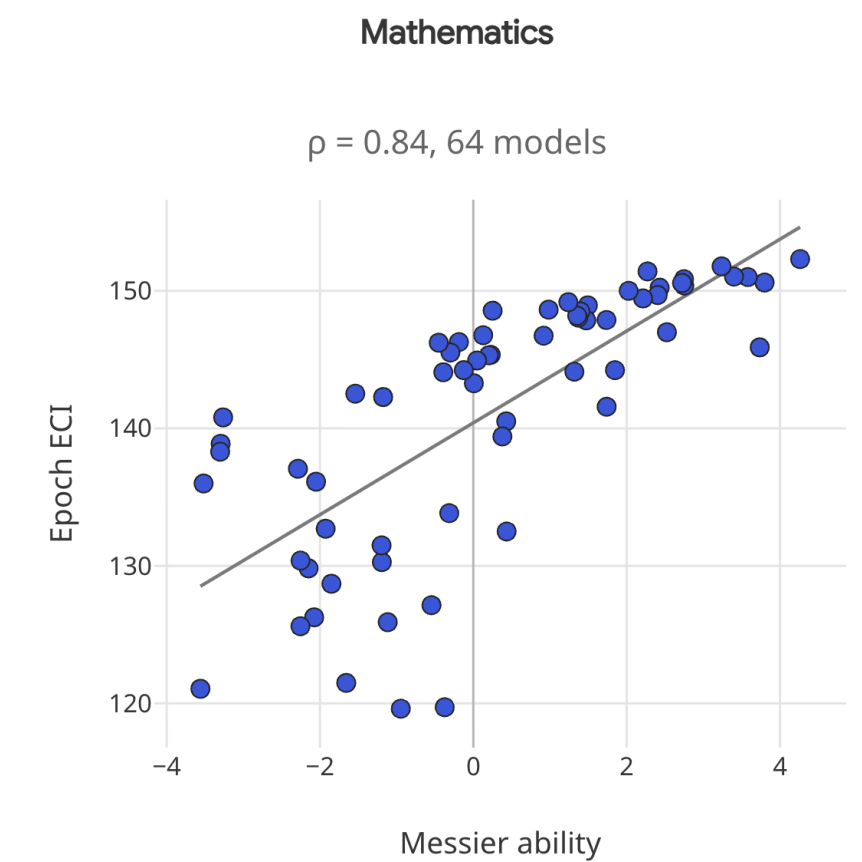
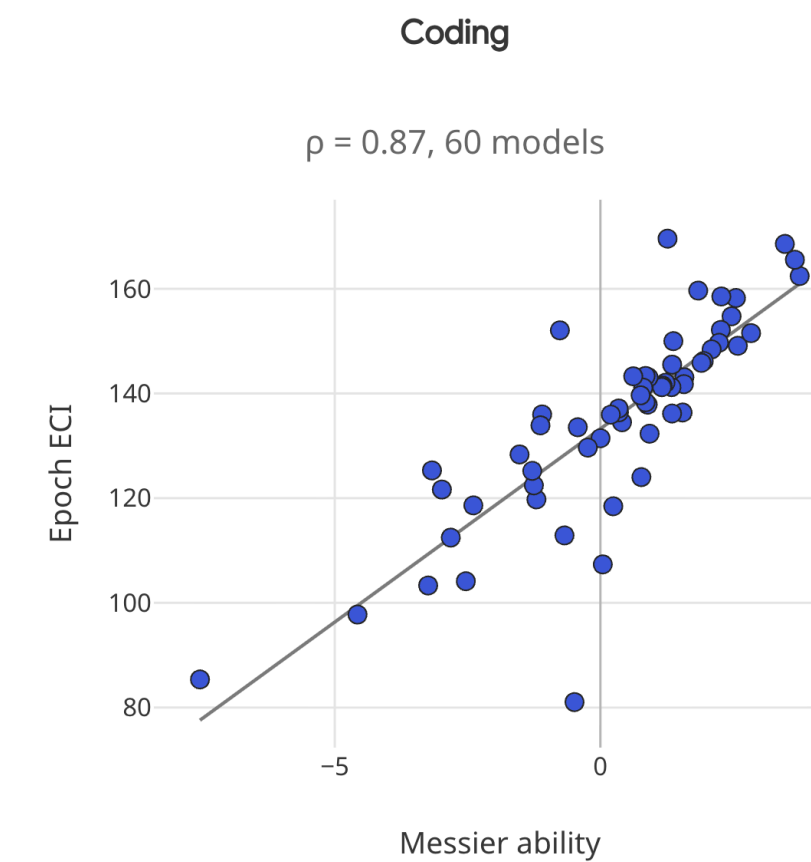
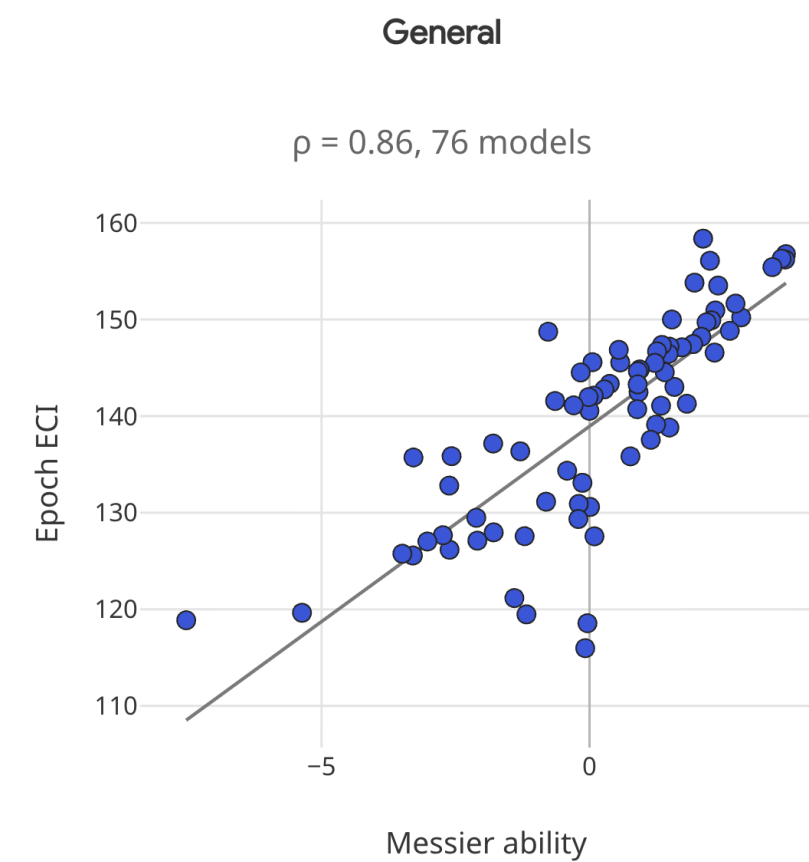
- **Setting:** tasks scored via multiple verifiers & all-pass
- **Method:** rescore the same trials with soft (fraction of verifiers passed) and $\geq 50\%$ threshold rules
- **Result:** HarveyAI-Lab: 0.2% of trials pass all-pass, 29.8% pass $\geq 50\%$, agent ranks shift ($\rho = 0.67$)
- **Takeaway:** report verifier results alongside trial results (aggregates)

Benchmark	All-pass	Soft	$\geq 50\%$	Near-pass	ρ (soft, all-pass)
HarveyAI-Lab	0.002	0.288	0.298	0.080	0.667
τ^2 -bench	0.479	0.754	0.809	0.113	0.857
TheAgentCompany	0.151	0.340	0.352	0.011	0.986

Analysis & Applications

Capability measurements (IRT)

- **Setting:** model ability from MESSIER vs. Epoch ECI (general, coding, math), and across NAICS industries
- **Method:** 1PL IRT (Rasch) with best scaffold per model-benchmark, Spearman ρ on matched models
- **Result:** $\rho = 0.86$ general, 0.87 coding, 0.84 math vs. ECI
- **Takeaway:** open data closely corresponds to ECI rankings, offering a form of validation for the data



Analysis & Applications

Task difficulty prediction

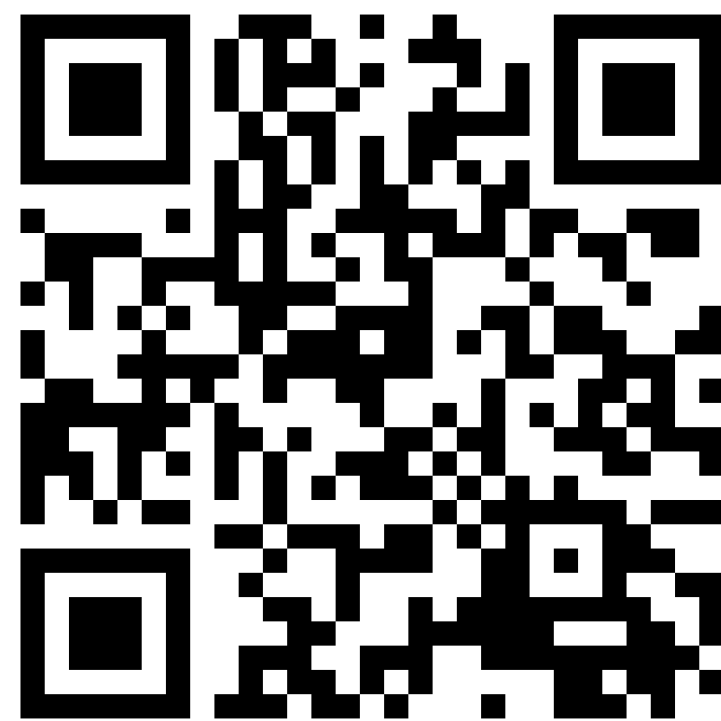
- **Setting:** 9,299 tasks with IRT difficulty β (raw, and z-scored within benchmark)
- **Method:** RidgeCV on bge-base-en embeddings of the instruction vs. all task metadata, 5-fold CV
- **Result:** z-scored β : $\rho = 0.30$ vs. -0.08 for the benchmark-mean baseline
- **Takeaway:** task descriptions carry difficulty signal, though results are still weak

Target	Input	Model	Baseline
β (raw)	instruction	0.484 ± 0.008	0.465
	all	0.508 ± 0.009	0.465
β (z-scored)	instruction	0.263 ± 0.015	-0.081
	all	0.298 ± 0.011	-0.081

Conclusion

- MESSIER includes 30 benchmarks, 745 agents, 11,891 tasks and 74,263 verifiers in one place, with more on the way
- As expected, keeping verifier results retains useful information
- Corresponds with closed-source capability measures ($\rho \approx 0.86$ vs. Epoch ECI)
- Allows for interesting analysis, e.g. slicing by certain occupations
- Open and extensible: contribute benchmarks, models and runs at github.com/Andromede-AI/messier

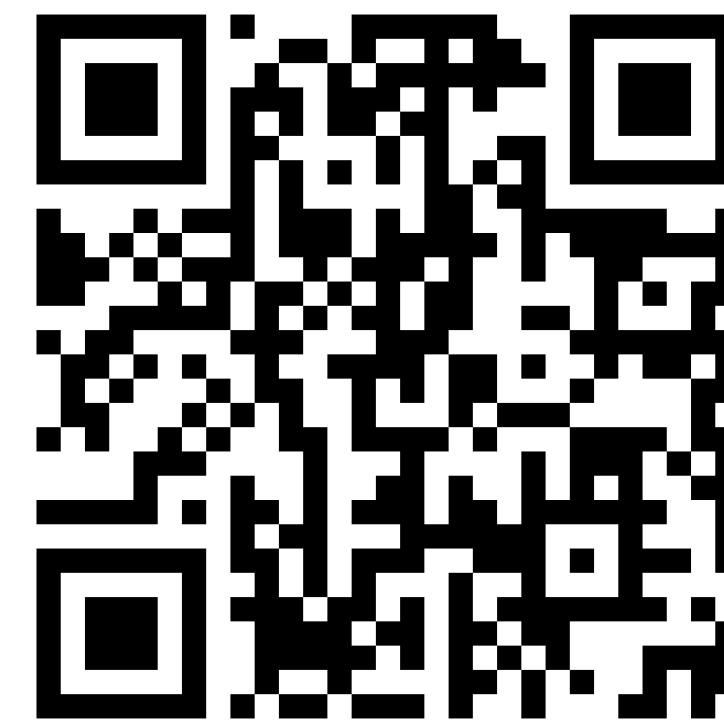
Thank you!



 **GitHub**
code



 **Hugging Face**
dataset



 **Project page**
more info

Krsteski, S., Meyer, C., Allegre, G., O'Halloran, T., & Sallinen, A. (2026). Messier: A High-Resolution Corpus for Cross-Benchmark Agent Evaluation. *arXiv preprint arXiv:2607.25891*.